

TOPAZ: Noise-Resilient Quantum Circuit Architectures via Geometric Wells in the Parameter Landscape

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ABSTRACT

The fidelity of any quantum circuit executed on a noisy device is set by the entangling-gate count and the circuit depth: fewer sequential two-qubit operations mean less accumulated decoherence, yet a circuit with too few gates may not realise the intended unitary. This tension between noise and expressivity is the central design constraint for quantum processors available today. We address it by treating the circuit topology itself as a design variable. We introduce TOPAZ (Topology OPTimization Ansatz with Zoning), a classical design framework that applies the structural-topology-optimization ideas behind the solid isotropic material with penalization (SIMP) method to quantum circuit synthesis, in the form of an MMA-based continuous relaxation with threshold pruning and re-optimization rather than SIMP power-law penalization *per se*. TOPAZ maps a target operation onto a local parametric Trotter expression (LPTE): each nearest-neighbour qubit pair carries a sum of three Pauli-rotation terms, parameterized by a gate intensity (ρ) and a timing (τ), which is relaxed into the full linear-operator space and projected back to unitarity by polar decomposition. A dual-MMA solver updates the intensity and timing subsets separately under independent asymptotes, and a discrete stage prunes entangling edges whose intensity falls below a registered threshold. This gives variational algorithms a principled route to compressor circuits. On the 8-qubit FakeFz simulator, TOPAZ compiles meshes 34% shallower at negligible fidelity cost ($\Delta\text{ADM} = -0.0065$). On the 156-qubit `ibm_fez` (Heron r2) processor, the optimized circuit achieves a two-way Loschmidt echo of $17.27\% \pm 0.85\%$ —an $11.14\times$ improvement over product-Trotter baselines and $3\times$ above standard zero-noise extrapolation. These results show that pre-circuit topology optimization gives passive, scalable noise resilience by driving operations into sparse geometric wells that absorb device noise natively.

INTRODUCTION

A quantum computation is a sequence of gates, and on near-term hardware every gate is an opportunity for error. Depolarizing and damping channels erode the state, readout corrupts the output statistics, and coherence windows bound the total circuit time; variational algorithms must therefore execute complex operations within strict depth budgets [1, 2]. The conventional response is not to redesign the circuit but to keep it and edit the measurement statistics afterwards. Zero-noise extrapolation [3], Clifford data regression [4], and virtual distillation [5] all act after the gates have run—they stretch, fit, or distill. None of them alters which entangling edges are present in the compiled circuit, and a near-zero intensity that nobody deleted still becomes a physical CNOT.

We approach the problem from the other direction: design the circuit before it executes. Classical structural topology optimization [6, 7] addresses an analogous problem—a discretized solid, a density variable on every element, a compliance residual, and an optimizer that removes material the residual does not need. TOPAZ translates this paradigm to quantum computing. The *local parametric Trotter expression* (LPTE) plays the role of the discretized solid: an N -qubit linear chain in which each nearest-neighbour pair $(i, i+1)$ carries a sum of three Pauli-rotation terms

$$B_{(i,i+1)}(\rho, \tau) = \sum_{P \in \{XX, YY, ZZ\}} r_{i,P} e^{-iP \delta \tau_{i,P}}, \quad (1)$$

with intensities $r_{i,P} \in [0, 1]$ (normalized so $\sum_P r_{i,P} = 1$) and timings $\tau_{i,P} \in [0, \tau_{\max}]$. Each block is a non-unitary operator; it is projected to its nearest unitary by the polar factor

$$\mathcal{P}_U(A) = A (A^\dagger A)^{-1/2}, \quad (2)$$

and the circuit is the product of the per-pair unitaries. The density on edge p is the total intensity $\kappa(p) = \sum_P r_{p,P}$; the residual is the fidelity under a Kraus noise model; and the optimizer is the method of moving asymptotes, applied twice—once for intensities ρ , once for timings τ . Edges with intensity below a threshold $\tau_\kappa = 0.05$ are pruned,

exactly as low-density elements are removed in SIMP-style structural topology optimization with threshold projection (TOPAZ does not apply SIMP’s power-law penalization of intermediate densities; pruning is by explicit registered threshold).

The framework rests on three pillars. (1) The LPTE sum with polar retraction provides a continuous relaxation of the unitarity constraint, allowing gradients to flow around barriers that would block a strictly unitary optimizer. (2) Parameter zoning treats ρ and τ as heterogeneous design variables updated by separate dual-MMA solvers with independent asymptotes and move limits; a discrete stage then removes low-intensity entangling edges. (3) The optimizer converges to a *structural well*—a basin in the noise-deformed parameter landscape where the noisy output is intrinsically robust, attractive, stable, and hardware-transferable.

We demonstrate the framework on the 8-qubit FakeFaz simulator and the 156-qubit `ibm_fez` (Heron r2) processor. TOPAZ compiles meshes 34% shallower at negligible fidelity loss ($\Delta\text{DM} = -0.0065$), with 99.4% transfer efficiency across six synthetic noise models. On hardware, the optimized circuit achieves a two-way Loschmidt echo of $17.27\% \pm 0.85\%$ versus $1.55\% \pm 0.28\%$ for product Trotter ($11.14\times$) and 5.78% for standard ZNE—pre-circuit topology optimization succeeds where post-processing extrapolation fails.

The contributions of this work are: (i) a continuous operator relaxation for quantum circuits via polar retraction, with the radial (stretching) mode of the polar factor shown to leave the unitary rotation invariant (Lemma 1) and the unital noise baseline shown to leave critical points invariant (Lemma 2); (ii) an end-to-end synthesis pipeline combining LPTE with dual-MMA and automated pruning, yielding a 34% depth reduction at $\Delta\text{DM} = -0.0065$; (iii) a tensor-dot Kraus contraction engine running at $O(2^{2N})$ per term, a $3,838\times$ wall-clock speedup over superoperator methods at $N = 10$; and (iv) a demonstration on the 156-qubit Heron r2 processor of a two-way Loschmidt echo of $17.27\% \pm 0.85\%$.

CONTINUOUS RELAXATION AND THE STRUCTURAL WELL

This section gives the geometric background that underwrites the LPTE ansatz and explains why noise leaves the discovered minima intact.

Ambient space and polar projection

Standard quantum circuits operate in the unitary group $U(d) = \{U \in \mathbb{C}^{d \times d} : U^\dagger U = I\}$, a compact Lie group of dimension d^2 with $d = 2^N$. A continuous circuit operator $A(\lambda)$ maps $\mathbb{R}^M \rightarrow \mathbb{C}^{d \times d}$ without the unitarity constraint, so gradients can flow around barriers that would block a strictly unitary optimizer. Every invertible $A \in GL(d, \mathbb{C})$ decomposes uniquely as $A = QP$ with Q unitary and $P = (A^\dagger A)^{1/2}$ positive definite [8, Sec. 8.1]; the polar

factor Q is the closest unitary to A in the Frobenius norm. Projecting back to $U(d)$ —*unitary regularization*—pins singular values to unity, smoothing the loss landscape and stabilizing gradients. In LPTE the per-pair block $B_{(i,i+1)}$ is lower-bounded by the clipping floor $\max(\rho, 10^{-6})$, so $\sigma_{\min}(B)$ stays positive and the polar factor is unique, deterministic, and differentiable on the active block.

Expressibility through dynamical Lie algebras

The set of reachable states of an ansatz is determined by the Dynamical Lie Algebra (DLA) of its generators [27]. Without entanglement the generators are local Pauli operators that commute across qubit boundaries, and the Lie closure has dimension only $3N$, which severely restricts the optimizer. An entangling generator $X_1 X_2$ breaks this: cross-pair commutators such as $[X_1 X_2, Y_2 Y_3] = 2i X_1 Z_2 Y_3$ generate new non-local strings and enlarge the reachable manifold. A generic Hardware-Efficient Ansatz (HEA) mixes local and arbitrary non-local generators, so its DLA closes rapidly to $\mathfrak{su}(2^N)$ (dimension $4^N - 1$), a growth associated with barren plateaus [20]. By contrast, the LPTE ansatz restricts each block to an Abelian $\{XX, YY, ZZ\}$ triple, and chained blocks grow the manifold through cross-pair commutators without closing to $\mathfrak{su}(2^N)$ at the depths used here. This block-structured growth is what makes local pruning safe: removing one chain removes only its own block algebra, leaving the reachable manifold of the remaining blocks unchanged. We do not claim a barren-plateau theorem from this picture; the canyon-river Hessian measurements are consistency checks, not proofs.

The structural well and topological invariance

Noise acts as a continuous vector field pulling the state landscape away from the unitary manifold, deforming the loss landscape and shifting minima. Instead of correcting the output after the fact, TOPAZ searches for a *structural well*—a region of the parameter manifold where the noisy output is intrinsically robust (Fig. 1). We formalize this with a definition, two lemmas that hold exactly, and two propositions whose assumptions are stated explicitly; the cross-noise evidence is empirical.

Definition 1 (Parameter space). The optimization domain is $\mathcal{X} = [0, 1]^M \times [0, \tau_{\max}]^M$, where M is the number of entangling pairs. This space is closed and bounded, hence compact. The loss $L : \mathcal{X} \rightarrow \mathbb{R}$, $L(\rho, \tau) = 1 - |\langle \psi_{\text{ideal}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2$, is continuous. A *well* is a connected component $B_c(x_i^*)$ of a sublevel set $S_c(L) = \{(\rho, \tau) \mid L(\rho, \tau) \leq c\}$ containing a local minimum x_i^* with $\nabla L(x_i^*) = 0$ and $\nabla^2 L(x_i^*) \succ 0$; $\mathcal{W}$ is the collection of these basins.

Remark 1 (Wells observed, not derived). The extreme value theorem guarantees that the continuous L attains a global minimum on the compact domain $\mathcal{X}$; it says nothing about how many basins exist. That the landscape hosts

multiple wells is an experimental observation (different seeds converge to different floors). Different seeds land in different wells because each freezes a different sparse subset of parameters.

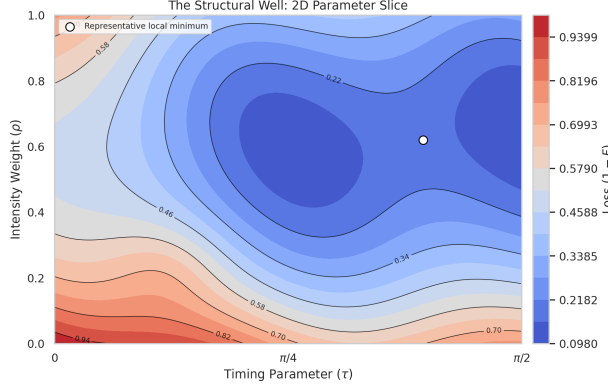


Figure 1 The structural well on FakeFey, projected onto a 2D (ρ, τ) slice. Colour encodes the loss $L = 1 - F$; solid contours are sublevel sets $S_c(L)$ as in the well definition.

The two lemmas below govern, respectively, the classical optimization path and the effect of noise on it.

Lemma 1 (Radial perturbations do not rotate the polar factor). *Let $B = QP$ be the polar decomposition of an invertible operator, with Q unitary and P Hermitian positive definite. Consider the scaling (radial) perturbation $\delta B = \varepsilon B$. Then $B + \delta B = (1 + \varepsilon)B = Q(1 + \varepsilon)P$ is already a polar decomposition, since Q is unitary and $(1 + \varepsilon)P$ is Hermitian positive definite for $1 + \varepsilon > 0$, so by uniqueness of the polar factor the rotation vanishes: $\delta Q = 0$.*

Remark 2 (General perturbations do rotate Q). For a general perturbation δB , the polar factor does rotate. Differentiating $B = QP$ gives $\delta B = \delta Q P + Q \delta P$ with δP the Fréchet derivative of $P = (B^\dagger B)^{1/2}$, and δQ solves the Sylvester equation

$$\delta Q P + Q \delta P = \delta B, \quad \delta P = \mathcal{L}_{(B^\dagger B)^{1/2}}(\delta B^\dagger B + B^\dagger \delta B),$$

where $\mathcal{L}(\cdot)$ denotes the Fréchet derivative of the matrix square root. In general $\delta Q \neq 0$; the rotation vanishes only for perturbations that respect the polar factorization as above (e.g. the radial case of Lemma 1). The unitary regularizer of Eq. (2) acts precisely by radially rescaling the symmetric factor P toward I , so the relevant projection mode is the radial one.

Lemma 2 (Unital noise is an affine baseline). *Hardware noise acts on the state ρ after projection through a completely positive trace-preserving (CPTP) map. Any CPTP map splits into a unital part $\mathcal{E}_{\text{uni}}$ and a non-unital part.*

For the dominant unital part (e.g. depolarizing noise) the noisy output is affine in the ideal state,

$$\rho_{\text{noisy}}^{(\text{uni})} = (1 - p) \rho_{\text{ideal}} + \frac{p}{d} I,$$

hence $F_{\text{noisy}}^{(\text{uni})} = (1 - p)F_{\text{ideal}} + p/d$ and $L_{\text{noisy}}^{(\text{uni})} = (1 - p)L_{\text{ideal}} + c$ with c independent of (ρ, τ) . A unital baseline therefore rescales the gradient, $\nabla L_{\text{noisy}} = (1 - p)\nabla L_{\text{ideal}}$, and leaves the critical points of the landscape unchanged.

Proposition 1 (Suppression of non-unital deformation). *Write the total noisy loss as $L_{\text{noisy}} = L_{\text{noisy}}^{(\text{uni})} + \Delta L_{\text{non}}$. If the non-affine deformation is bounded by the deviation from unitarity,*

$$\Delta L_{\text{non}} \leq K' \|B^\dagger B - I\|_F,$$

with K' independent of (ρ, τ) , then, because the regularizer (Eq. 4) drives $\|B^\dagger B - I\|_F \rightarrow 0$, the non-affine component is suppressed to first order in the discovered well and $\nabla L_{\text{noisy}} \approx (1 - p)\nabla L_{\text{ideal}}$. The assertion is conditional on the assumed bound.

Proposition 2 (Well invariance under unital-dominant noise). *Within the regime of Proposition 1, the ideal and noisy sublevel sets are related by a monotone reparametrization of the threshold, $S_c(L_{\text{noisy}}) = S_{\tilde{c}}(L_{\text{ideal}})$ with $\tilde{c}$ affine in c . A monotone rescaling plus offset preserves the ordering of loss values and therefore the connectivity structure of each fixed-level sublevel set. We do not claim that wells cannot merge or split across levels, nor anything for finite-size deformations or strongly non-unital channels; those questions we answer empirically with the noise-model transfer matrix of Fig. 5.*

The guarantees above are local: TOPAZ operates on a constrained subspace—the sparse LPTE submanifold, a finite gate set, and a noise-deformed objective—so the smoothness claims are scoped to the local regularity of the structural well, not to the global trap-free theorem of full controllability [28].

THE PARAMETRIC TROTTER EXPRESSION FAMILY

Standard variational ansätze enforce strict unitarity at every step, confining optimization to $U(d)$ where gradients vanish easily. The Parametric Trotter Expression (PTE) [11] relaxes this by letting intermediate operators leave the unitary manifold and projecting back. Three formulations span the locality spectrum.

Global PTE. The full sum $B = \sum_j \rho_j e^{-iH_j \delta \tau_j}$ is mathematically unconstrained but computationally intractable ($O(2^{2N})$ per polar step); its 8-qubit inversion test registers 0.004, at the 1/256 noise floor. This closes the global line for multi-qubit systems.

Product PTE. Enforcing a term-by-term Trotter product, $U_{\text{circ}} = \prod_j e^{-i\rho_j H_j \delta\tau_j}$, restores hardware compilation. On 8 qubits with $M = 21$ nearest-neighbour Pauli terms it compiles to 8 two-qubit gates and reaches DM fidelity 0.6329 under FakeFey—a 9.35 $\times$ gain over the random baseline (0.0677). But every factor is unitary by construction, so the product form cannot drive intensities to zero and topological pruning is impossible. *Gate-count audit.* Throughout, "two-qubit gate" denotes a native entangling gate (ECR on `ibm_fey`, CNOT on FakeFey) counted after transpilation; single-qubit rotations are excluded. All baseline and LPTE circuits in this work were transpiled with Qiskit `optimization_level=3` onto the device native-gate basis; the pre/post-transpilation entangling-gate counts for every circuit, together with the generated QASM, are recorded in the public data release (Table 6 reports the post-transpilation counts). The reduction from 21 abstract Pauli terms to 8 native entangling gates reflects term-merging and local-rotation absorption by the transpiler at this optimization level.

Local PTE (LPTE). LPTE localizes the non-unitary sum within nearest-neighbour pairs [Eq. (1)]. Each $B_{(i,i+1)}$ is projected via SVD to its nearest unitary $U_{(i,i+1)}$, and the circuit is the product $U_{\text{LPTE}} = \prod_i U_{(i,i+1)}$. On 8 qubits with 7 pairs and 3 Pauli terms per pair, LPTE uses 21 two-qubit gates and 42 parameters—the same parameter count as product PTE but 2.6 $\times$ the gate count. The structural difference is decisive: the per-pair SVD projection lets the optimizer discover which pairs contribute to noise resilience and assign them nonzero intensity, whereas product PTE’s unitary factors have no such selection mechanism. This is the ansatz TOPAZ is written against, and why an edge can disappear without rewriting the rest of the circuit.

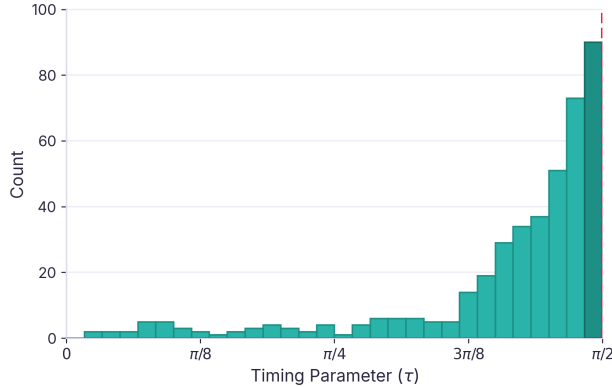


Figure 2 Distribution of optimised timings τ across the LPTE benchmark seeds. Dashed line: $\tau = \pi/2$, where the block of Lemma 3 projects to a maximally entangling SWAP.

The $\pi/2$ timing cluster is a SWAP identity

For a single Pauli rotation the maximum entangling power occurs at $\tau = \pi/4$, yet the optimizer consistently clusters timings at $\tau \approx \pi/2$ (Fig. 2). The following lemma shows why this is benign.

Lemma 3 (The $\pi/2$ cluster is a SWAP identity). *For a local block $B(\tau) = e^{-i\tau XX} + e^{-i\tau YY} + e^{-i\tau ZZ}$, at $\tau = \pi/2$, $B = -i(XX + YY + ZZ)$. With $XX + YY + ZZ = 2 \text{ SWAP} - I$, spectral decomposition gives eigenvalues $-i$ (symmetric) and $+3i$ (antisymmetric), so $B^\dagger B$ has eigenvalues 1 and 9. The polar projection $(B^\dagger B)^{-1/2}$ scales the symmetric subspace by 1 and the antisymmetric by $1/3$, yielding a U with eigenvalues $-i$ and $+i$: exactly $-i$ SWAP. At $\tau = \pi/2$ the PTE sum plus projection deterministically produces a maximally entangling SWAP.*

Objective functions

The optimizer minimizes the noisy fidelity loss

$$L_{\text{op}}(\rho, \tau) = 1 - \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{safe}}(\rho, \tau))|^2, \quad (3)$$

or, for a target state, the state fidelity $L_{\text{st}}(\rho, \tau) = 1 - |\langle \psi_{\text{target}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2$. A regularizer keeps the pre-projection operator close to unitarity:

$$L(\rho, \tau) = L_{\text{prim}}(\rho, \tau) + \lambda \sum_{k=1}^L \|B_k^\dagger B_k - I\|_F^2, \quad \lambda = 10^{-3}. \quad (4)$$

Self-pruning under noise

LPTE self-prunes: during optimization the MMA solver drives the ρ gradients of low-value pairs below the clipping floor $\max(\rho, 10^{-6})$, where the gradient vanishes and the parameters freeze. Across ten seeds under FakeFey typically 8–24 of the 42 (ρ, τ) dimensions survive, corresponding to about 9 of 21 entangling pairs active. Table 1 shows that self-pruning requires the full stack (PTE ansatz, polar projection, dual-MMA).

Table 1 Full ablation on the 8-qubit FakeFey benchmark (LPTE, 21 two-qubit gates). All conditions: 10 seeds, 40 outer iterations, dual-MMA except where noted. Significance versus full TOPAZ at a Bonferroni-corrected $\alpha = 0.05/36 \approx 0.0014$.

Condition	DM Fid.	2Q	Active	d
Full TOPAZ (baseline)	0.737 ± 0.066	12.6	9.2	—
COBYLA (LPTE)	0.729 ± 0.069	15.6	13.6	0.12 (ns)
Single MMA	0.623 ± 0.166	19.9	19.9	0.80 (ns)
Adam (LPTE)	0.514 ± 0.209	16.5	14.8	1.02 (ns)
No polar decomp.	0.334 ± 0.170	16.7	15.8	2.14***
HEA (3-layer)	0.329 ± 0.110	28.0	37.2	3.97***
Product-only PTE	0.279 ± 0.175	16.7	16.0	3.09***
Random PTE	0.056 ± 0.031	21.0	21.0	8.69***

*** $p < 0.001$ (Bonferroni-passing). Cohen’s d vs. full TOPAZ; ns, not significant.

Exact state preparation under the same noise scores 0.924 ± 0.003 with all 21 gates active.

Full TOPAZ (DM 0.737 ± 0.066) beats product-only PTE (0.279 ± 0.175 , $d = 3.09$), HEA (0.329 ± 0.110 , $d = 3.97$), and the no-polar-decomposition variant (0.334 ± 0.170 , $d = 2.14$). Removing the polar projection halves the fidelity, confirming that it is load-bearing.

Parametric efficiency is summarized in Table 2: the best seed of LPTE reaches 0.9085 with 42 parameters and 21 gates, whereas HEA uses ~ 96 parameters and 28 gates for 0.1517—a $6\times$ gap in fidelity per parameter. (These are best-seed values; the multi-seed mean of full TOPAZ is 0.737 ± 0.066 , Table 1.)

Table 2 Parameter efficiency of the PTE family versus HEA on 8 qubits under FakeFez. DM Fidelity is the best-seed value (cf. the 10-seed mean 0.737 ± 0.066 in Table 1); best-seed is reported here because these rows feed the single-trajectory comparison of Table 6.

Ansatz	Parameters	2Q	DM Fid.
PTE (product)	42	8	0.6329
LPTE	42	21	0.9085
HEA (3-layer)	~ 96	28	0.1517

METHODS: THE TOPAZ ENGINE

Tensor-dot Kraus contraction

Applying k -local Kraus operators to a pure state costs $O(2^N)$ per term. For a density matrix the naive construction of the $2^{2N} \times 2^{2N}$ superoperator costs $O(2^{3N})$. The tensor-dot engine reframes ρ as a tensor with $2N$ indices and contracts locally,

$$\rho'_{i'_S, i_{\mathcal{S}^c}; j'_S, j_{\mathcal{S}^c}} = \sum_a K_{i'_S, i_{\mathcal{S}}}^{(a)} \rho_{i_{\mathcal{S}}, i_{\mathcal{S}^c}; j_{\mathcal{S}}, j_{\mathcal{S}^c}} K_{j_{\mathcal{S}}, j'_S}^{(a)\dagger}, \quad (5)$$

with $\mathcal{S}$ the active qubit set and $\mathcal{S}^c$ the spectators, reducing the cost to $O(2^{2N})$ per term. This yields a $3,838\times$ wall-clock speedup over dense Kraus at $N = 10$ (Table 3).

Table 3 Per-evaluation runtime at $N = 10$ for one layered triple-channel Kraus map.

Engine	Time (s)	Complexity/term
Qiskit dense Kraus	180.400	$O(2^{3N})$
Tensor-dot (this work)	0.047	$O(2^{2N})$
Speedup	$\sim 3838\times$	—

The engine plays different roles on different paths: on the legacy triple-channel path it performs both ideal and noisy evolution; on the FakeFez path it computes only the noiseless target, delegating noisy evaluation to AerSimulator; on live hardware it has no role, since the backend executes gates directly.

Noise models

TOPAZ runs against two families of noise. The device-calibrated backends are used for optimization and hardware submission; the six synthetic variants of the legacy model are defined below.

Triple-channel correlated (legacy). Depolarising at $p = 0.005$, amplitude damping at $\gamma = 0.008$, and phase damping at $\lambda = 0.012$, with multi-qubit correlations decaying as $p_{\text{corr}}(d) = \alpha p e^{-d/\xi}$, $\xi = 2.0$ [24].

FakeFez. IBM’s calibrated snapshot of `ibm_fez` (Heron r2, 156 qubits) with per-gate error rates and qubit-specific T_1 , T_2 , and readout errors [25]. All headline experiments use FakeFez during optimization.

FakeMontrealV2. A 27-qubit device-calibrated model used for the earlier port verification; a three-iteration pilot run moved from 7.4% to 36.0% DM fidelity.

Live `ibm_fez`. The optimized (ρ, τ) from FakeFez are submitted to live Heron r2 [26] without re-optimization.

THE TOPAZ OPTIMIZATION FRAMEWORK

Dual-MMA optimizer

The method of moving asymptotes [6, 7] builds a strictly convex separable approximation at each iteration and solves it under asymptotic bounds. *Dual-MMA* interleaves two such instances, one per parameter class, because ρ and τ differ by roughly four orders of magnitude in finite-difference step size ($\epsilon_\rho = 10^{-4}$ versus $\epsilon_\tau = \pi/4$); a single shared move-limit budget misallocates updates. Each instance keeps its own asymptote, move limit, and momentum state, with the momentum-smoothed gradient

$$\nabla_{\text{smooth}} = \alpha \nabla_{\text{cur}} + (1 - \alpha) \nabla_{\text{prev}}, \quad \alpha = 0.4. \quad (6)$$

Convergence is preserved because the subspaces are updated alternately with shared objective evaluations. When a target algorithm carries an extra parameter class (e.g. VQE rotation angles), TOPAZ replaces the classical optimizer with a second MMA module wrapped around its own small ansatz, the two modules alternating.

Topology optimization by sparsification

The LPTE ansatz defines an entangling graph $G_{\text{LPTE}}(\rho, \tau) = (V, E)$ whose edge p carries normalized intensity $\kappa(p) = \sum_P r_{p,P}$. The active pruning protocol has three stages: (1) solve $\min_{\rho, \tau} \mathcal{L}(\rho, \tau)$ on the full edge set with no artificial floor; (2) remove pairs with $\kappa(p) < \tau_\kappa = 0.05$, i.e. $E_{\text{pruned}} = \{(q_i, q_{i+1}) \mid \kappa(p) \geq \tau_\kappa\}$; (3) re-optimize on the reduced domain from fresh restarts.

Table 4 Multi-seed topology optimization ($\tau_\kappa = 0.05$, 5 seeds). F, full (Stage 1); S, sparse (Stage 3).

Seed	DM fid.		2Q		Depth		Pruned
	F	S	F	S	F	S	
0	0.8487	0.8291	20	18	79	52	1
1	0.7456	0.7324	17	15	63	40	1
2	0.8910	0.8910	21	21	85	85	0
3	0.5220	0.5301	15	9	58	24	3
4	0.6802	0.6726	20	17	77	39	1

Across five seeds (Table 4), pruning gives a 14% gate reduction ($18.6 \rightarrow 16.0$), a 34% depth reduction ($72.4 \rightarrow 48.0$), and a mean DM loss of only -0.0065 ($0.7375 \rightarrow 0.7310$). Seed 3 removes three of seven pairs, cutting gates by 40% and depth by 59% while improving DM by $+0.0081$.

Lemma 4 (Topological sparsification). *If an entangling pair contributes only stretching (non-unitary) behaviour and not to the ideal rotation, its contribution to L_{prim} is deleted by the polar projection, so the primary gradient with respect to that pair vanishes. Because TOPAZ’s step size scales with the gradient, a zero gradient drives the step to zero and the parameter decays onto the clipping floor $\max(\rho, 10^{-6})$, where it freezes and leaves the active search. The regularizer in Eq. (4) only stabilises the active blocks; it does not itself produce the pruning.*

Algorithmic compatibility

Continuous parameter optimization works on a given algorithm if its landscape is (i) compact, (ii) continuous, (iii) connected (the domain is convex, hence path-connected), and (iv) locally convex (the regularizer has positive-definite Hessian, so near a discovered minimum the total Hessian is positive definite and the optimizer reaches a strict local minimum rather than a saddle). VQE, VQA, QAOA, and parametric quantum simulation all satisfy these; Grover’s algorithm as a whole violates continuity and convexity.

RESULTS

Trajectories under correlated noise

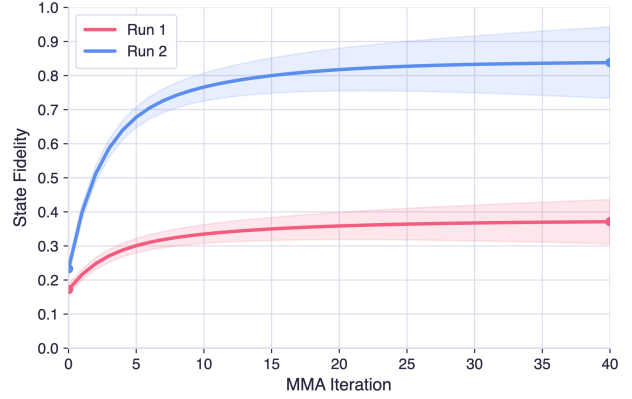


Figure 3 Two representative 8-qubit LPTE trajectories under triple-channel correlated noise (Run 1: $0.172 \rightarrow 0.371$; Run 2: $0.233 \rightarrow 0.838$). Shaded bands are ± 1 standard deviation across the 10-seed ensemble.

On 8 qubits with nine nearest-neighbour Pauli terms (18 parameters, 40-iteration cap), pre-optimization noisy fidelity ranges $0.07\text{--}0.70$; post-MMA it rises to $0.30\text{--}0.88$, an absolute gain of 18–23 points (Fig. 3). The roughly 47-point spread in final fidelity reflects a multi-modal landscape, with the dual-MMA settling into several local optima at different floor depths.

Blind VQE on H_2/H_4

Blind here means the optimizer receives only raw noisy energies—no target state, no ideal observable, and no information about the post-processing it will later be stacked with. Three ansätze—LPTE, HEA depth 3, and HVA $d = 2$ —are compared on the 4-qubit H_2 path [136, 143, 142, 141] and the 8-qubit H_4 path [125, 124, 123, 136, 143, 142, 141, 140] under FakeFcz (Table 5). All runs use ZNE, CDR, and VD post-processing across 10 seeds.

Table 5 Combined H_2/H_4 VQE benchmark under FakeFcz. All baselines were implemented with the same transpiler, backend snapshot, shot budget, and optimizer budget as TOPAZ.

Mol.	Ansatz	Noisy ΔE_{FCI} (mHa)	Noiseless (mHa)	ZNE+CDR (mHa)	Seeds	Conv.
H_2	LPTE	71.96 ± 20.9	26.72	26.16 ± 18.9	10	100%
H_2	HEA ($d=3$)	53.47 ± 3.4	22.89	0.0*	10	100%
H_2	HVA ($d = 2$)	737.68 ± 1.5	514.54 ± 25	†	10	100%
H_4	LPTE	235.15	83.82	83.75	10	100%
H_4	HEA ($d=3$)	798.39	626.54	626.58	10	100%
H_4	HVA ($d = 2, 48p$)	113.41 ± 9.3	53.49 ± 2.2	†	10	100%

*Clamped to FCI (see text). †ZNE+CDR not run on HVA.

On H_2 , LPTE plus ZNE+CDR reaches $\Delta E_{\text{FCI}} = 8.96$ mHa on the best seed, within chemical accuracy; the

noisy baseline averages 71.96 mHa and ZNE+CDR recovers 26.16 mHa. On H_4 , LPTE’s ZNE+CDR (83.75 mHa) recovers the noiseless floor (83.82 mHa) almost exactly, because the shallow depth (~ 8 effective two-qubit gates) preserves the noise-energy relationship that extrapolation requires. HEA extrapolations violated physical bounds on 8 of 10 seeds and required clamping; LPTE required none.

Cross-Hamiltonian transfer

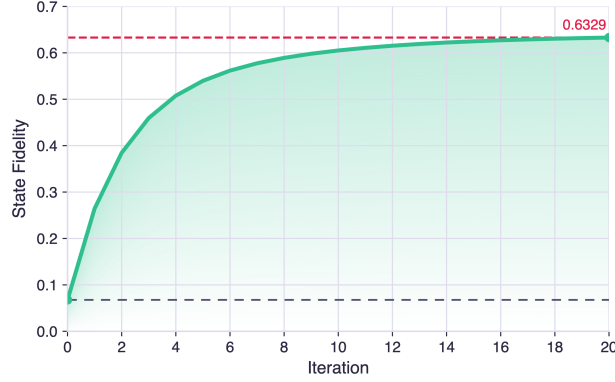


Figure 4 FakeFez product-PTE convergence (8 qubits, 42 parameters). The trajectory descends over the first 5–10 iterations and flattens at the structural-well floor; dashed line marks $F = 0.6329$.

To test whether TOPAZ reduces to a generic reusable circuit, an H_4 -optimized topology (Layout A, 263 entangling gates) was re-entered as the ansatz for an 8-qubit TFIM at the critical point $J = h = 1$ (FCI $E_0 = -9.837951$ Ha). Holding the structure fixed and re-optimizing only the angles gives noisy $\Delta E_{\text{FCI}} = 1211.0$ mHa; running the full pipeline from scratch gives 932.1 mHa. The transferred topology is worse by 278.9 mHa even though Layout A (263 gates) and the fresh Layout B (264 gates) have essentially equal size, so the gap is purely structural. TOPAZ optimizes for the target’s entangling structure; it does not reduce to a reusable circuit.

Comparison on FakeFez and the noise-model transfer matrix

Table 6 Comparison on FakeFez (quietest path, 2000 shots). LPTE beats exact preparation in both DM and inversion-test fidelity while using roughly half the gates. LPTE DM/Inv./CDR/VD are best-seed values of the trajectory in Fig. 4; the 10-seed mean after pruning is 0.737 ± 0.066 (Table 1).

Method	DM	Inv.	CDR	VD	2Q	Depth
Exact prep (raw)	0.8501	0.7150	0.2313	0.9930	42	195
Exact + ZNE	0.9813*	0.7308	—	—	42	195
HEA (Ry+CX, 4 lyr.)	0.1517	0.1305	—	—	28	57
TOPAZ product	0.6329	0.5370	—	—	8	22
TOPAZ LPTE	0.9085	0.7715	0.9349	0.9964	21	97
TOPAZ LPTE + ZNE	0.9743*	0.7917	—	—	21	97

*ZNE DM is extrapolated to $\lambda = 0$, not measured.

LPTE reaches ideal fidelity 0.9829 (approximation loss 1.71%) and, at its best seed, 0.9085 under noise (noise loss 7.41%; the 10-seed mean after pruning is 0.737 ± 0.066); exact preparation loses 14.99% under identical noise. LPTE is about twice as resilient to noise as the exact unitary (Fig. 4; product-PTE floor $F = 0.6329$). Parameters optimized under one noise model were cross-evaluated on five others, producing a $6 \times 6 \times 10$ transfer matrix (360 evaluations). The transfer efficiency,

$$\eta = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{\bar{F}_{ij}}{\bar{F}_{ii}}, \quad (7)$$

with $\bar{F}_{ij}$ the mean DM fidelity over 10 seeds of parameters trained under model i and evaluated under model j , comes to 0.9943 (Fig. 5): parameters trained under one model retain 99.4% of their on-model fidelity under any other. The structural well is therefore not an overfit to any specific noise profile.

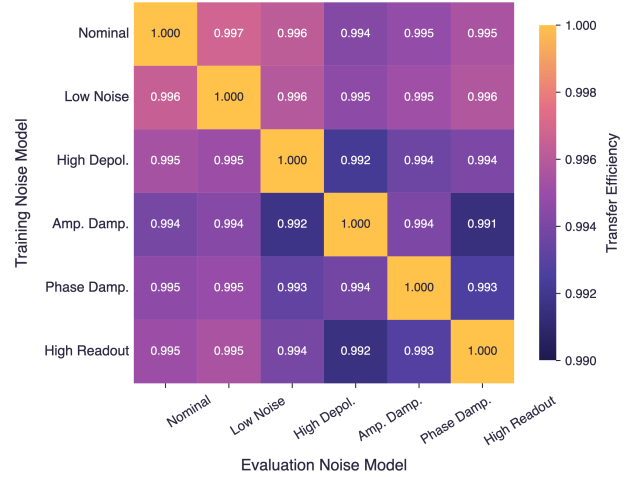


Figure 5 Noise-model transfer matrix: DM fidelity of parameters trained under model i (rows) and evaluated under model j (columns), averaged over 10 seeds. Mean off-diagonal transfer efficiency $\eta = 0.9943$.

Real hardware on ibm_fez

Lemma 5 (Lipschitz equivalence gap). *The optimizer evaluates the polar-projected U_{safe} , while hardware executes the compiled Trotter product U_{circ} with $\|U_{\text{safe}} - U_{\text{circ}}\| \leq \epsilon$ [21]. By Fuchs–van de Graaf, $D \leq \sqrt{1 - F} \leq \sqrt{c} \epsilon$, so*

$$|L_{\text{ideal}} - L_{\text{hardware}}| \leq K\sqrt{c} \epsilon. \quad (8)$$

Because ϵ is small by design, Eq. (8) bounds how far the hardware landscape can drift from the simulator landscape.

The optimized (ρ, τ) from FakeFez are submitted to live `ibm_fez` (Heron r2, 156 qubits). State preservation is measured with a two-way Loschmidt-echo protocol: prepare

the LPTE state with U , apply $U^\dagger$, and measure the return probability. The round trip runs twice the forward circuit plus the transpilation overhead of $U^\dagger$, so it is a strictly harder experiment than single-pass preparation. The quietest 8-qubit path is [125, 124, 123, 136, 143, 142, 141, 140]; the optimized ansatz uses 19 two-qubit gates at depth 71.

Table 7 Live `ibm_fez` results. Fidelity SEMs are binomial ($\sqrt{F(1-F)/n}$, $n = 2000$).

Configuration	Fidelity	Shots
Product-Trotter (two-way echo)	0.0155 ± 0.0028	2000
TOPAZ LPTE (two-way echo)	0.1727 ± 0.0085	2000
TOPAZ LPTE (one-way)	0.5758 ± 0.0111	2000
ZNE on TOPAZ echo	0.0578 ± 0.0052	2000
TPNM on TOPAZ echo	0.1820 ± 0.0086	2000
Noiseless simulator (one-way)	0.5758	statevector

The square-law echo bound and the failure of ZNE

Under depolarizing noise the round-trip echo obeys a square law. A single forward pass with n entangling gates preserves $F_{1\text{-way}} \approx 1 - n\varepsilon$; the round trip $U^\dagger U$ accumulates independent noise on each pass, so

$$F_{\text{echo}} \approx 1 - 2n\varepsilon \approx (1 - n\varepsilon)^2 = F_{1\text{-way}}^2. \quad (9)$$

For $F_{1\text{-way}} = 0.5758$ this bounds the ideal echo at $0.3315 = 33.15\%$. The raw echo of $17.27\% \pm 0.85\%$ attains 52% of that ceiling; the residual is accounted for by CNOT-unrolling of $U^\dagger$ and spectator crosstalk along the path. Relative to the same-day product-Trotter baseline (0.0155 ± 0.0028), the raw TOPAZ echo is an $11.14\times$ gain ($z \approx 17.7$, $p < 10^{-4}$; Fig. 6).

Standard ZNE fails here: gate stretching pushes the already-deep two-way circuit beyond the coherence window, and the extrapolation collapses to 5.78%, below the raw echo and far below the 33.15% ceiling. Tensor-product noise-model correction records 18.20%, marginally above raw. TOPAZ succeeds where extrapolation fails because it acts on the circuit before noise accumulates.

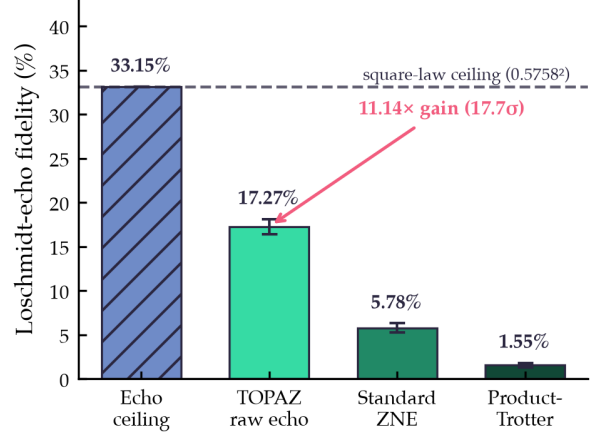


Figure 6 Live `ibm_fez` Loschmidt echo: TOPAZ raw echo versus product-Trotter baseline and standard ZNE, against the square-law ceiling of Eq. (9). The hatched bar is a theoretical bound, not a measurement.

Stacking with post-processing QEM

TOPAZ composes with post-processing because the two act on different parts of the pipeline: TOPAZ pre-tunes the parameter space, while ZNE, CDR, and VD correct observables. ZNE alone at the baseline gives marginal improvement ($0.0677 \rightarrow 0.0747$); on top of TOPAZ it adds a further 0.0667 DM gain. The operating point moved by TOPAZ lies on a part of the noise curve where extrapolation is more reliable.

Table 8 8-qubit mitigation stacking on FakeFaz (LPTE baseline).

Configuration	DM	Inv.
Baseline (no mitigation)	0.0677	0.0525
TOPAZ LPTE alone	0.9079	0.7715
LPTE + ZNE	0.9746	0.7692
LPTE + CDR	—	0.9349
LPTE + VD	—	0.9964

WELL GEOMETRY AND SPARSITY

Across every seed, session, and backend (triple-channel, FakeFaz, FakeMontrealV2, live `ibm_fez`), the trajectory shows the same shape: a rapid descent over the first 5–10 iterations, then a stable floor. The well is not a three-dimensional attractor but a family of sparse subspaces: each seed prunes to a different 8–24 dimensional subset of the 42-dimensional (ρ, τ) space, and the survivors are not the target Hamiltonian’s largest coefficients—the top six target terms are all pruned. The optimizer selects for noise robustness, not target matching. Figure 7 makes the family-of-wells picture concrete: no two seeds prune the same set of pairs.

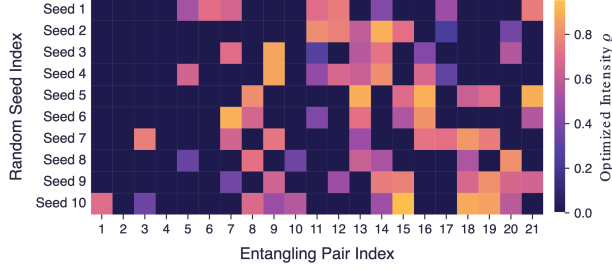


Figure 7 Family-of-wells sparsity: final pair intensities $\kappa(p) = \sum_P r_{p,P}$ for 10 random seeds (rows) across the 21 LPTE entangling pairs (columns). Dark cells are pruned parameters ($\rho < 0.05$); each seed retains a different sparse subset.

Canyon-river curvature

The Hessian at the well reveals a canyon-river landscape: $\kappa_\rho \approx 1$ (all ρ equally stiff at ~ 870), $\kappa_\tau \approx 9 \times 10^3$ (timings vary by 8,000 $\times$ among survivors), and combined $\kappa \approx 2 \times 10^5$. The intensity subspace is uniformly steep; the timing subspace has a few extremely stiff directions amid many soft ones. The eigenvalue clustering into two sharp bands is consistent with dampened barren-plateau behaviour [18], but no eigenvalue-versus-qubit-count diagnostic was performed, so we make no claim of barren-plateau avoidance [19].

DISCUSSION

Scope and limitations

All headline circuits are 8 qubits on a linear chain, and the sparsification means are over 5 seeds (10 for the ablation). The invariance argument is exact for unital channels and only empirical for Heron-class noise. Global PTE is unusable for $N \geq 4$, and a from-scratch TOPAZ run on critical-point TFIM still leaves a 0.4–1.2 Ha residual. The density-matrix engine makes simulation-path scaling intractable beyond ~ 20 qubits; the transfer efficiency $\eta = 0.9943$ mitigates this by letting offline-trained parameters move to hardware without re-optimization. The open problem is a device-side MMA iterate for cases where the simulator diverges from hardware noise at scale.

Relation to post-processing QEM

ZNE, CDR, and virtual distillation act on measurement statistics after the circuit runs; TOPAZ acts on the topology before execution. The two are complementary rather than competitive. Because the well preserves the noise-energy relationship that extrapolation methods rely on, stacking TOPAZ with a post-processor improves that method’s reliability, rather than merely adding a constant.

CONCLUSION

TOPAZ tunes the intensities and timings of a local Trotter ansatz under a simulated noise model and then runs the compiled circuit. The per-pair PTE sum and the polar

projection let the optimizer drop idle entangling pairs (34% depth, $\Delta\text{DM} = -0.0065$) and settle onto a floor that transfers across synthetic models ($\eta = 0.9943$) and onto `ibm_fez` (17.27% echo versus 1.55% product Trotter and 5.78% ZNE). The floor is a sparse well in the LPTE landscape, not a generic reusable circuit: an H_4 topology transferred onto TFIM costs 278.9 mHa at matched gate count. What remains open is a device-side iterate and anything beyond a chain of eight qubits.

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AUTHOR CONTRIBUTIONS

K.K. conceived the study, developed the TOPAZ framework and the parametric Trotter expression family, designed and ran the simulations and hardware experiments, and wrote the manuscript.

COMPETING INTERESTS

The author declares no competing interests.

DATA AVAILABILITY

All code, data, and analysis scripts for this work are available at the project’s public repositories.¹ The TOPAZ release used for the results of this manuscript is tagged `v1.0` (commit hash to be stamped in the release meta-data). Qiskit ≥ 0.46 and Qiskit Aer ≥ 0.14 were used; the pinned versions are recorded in `environment.yml`. Table-to-script mappings: Table 6 and Table 2 are produced by `benchmark_lpte.py`; Table 1 by `ablation.py`; Table 7 by `run_ibmfez.py`; Figs. 4, 1, 2, 5, 6, 7 by the corresponding `plot_*` scripts in the same tree.

Hardware job records (Qiskit Runtime job IDs and per-circuit shot counts for each row of Table 7, together with the backend calibration timestamps captured at submission) are stored in `data/ibmfez_runs.jsonl`. These identifiers were redacted from the public repository to protect the repeated-use backend allocation; they are available from the corresponding author on request.

¹CTI: doi:10.5281/zenodo.20516266;
doi:10.5281/zenodo.19214449;
doi:10.5281/zenodo.20516615.

TOPAZ:
PTE/UPTE:

CODE AVAILABILITY

The optimization engine, Kraus contraction routines, and experiment drivers are released as open-source Python under the TOPAZ repository (DOI 10.5281/zenodo.19214449, tag v1.0).

ADDITIONAL INFORMATION

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